

ORKDAL-THAMSHAMN, Norway

WMO: 012340

Lat: 63.3153N

Lon: 9.8494E

Elev: 4

StdP: 101.28

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.8	-15.7	-21.0	0.6	-18.8	-17.6	0.8	-15.2	11.2	0.7	9.9	1.4	4.4	210	0.682

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.7	23.1	16.3	20.8	15.3	19.2	14.8	17.4	21.1	16.3	19.5	15.4	18.4	4.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	11.3	19.1	14.7	10.5	17.8	13.8	9.9	17.1	49.1	20.9	45.7	19.7	43.0	18.4	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.1	8.0	7.1	-19.5	27.8	3.6	2.1	-22.0	29.3	-24.1	30.6	-26.1	31.7	-28.7	33.3
			-19.6	18.7	3.7	1.5	-22.3	19.8	-24.4	20.7	-26.5	21.6	-29.1	22.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-3.3	-2.3	1.0	4.6	8.8	12.0	14.9	13.8	10.5	5.4	0.2	-1.8
	DBStd	7.45	5.64	5.52	3.74	2.86	3.36	2.83	2.94	2.43	2.67	3.26	4.69	5.67
	HDD10.0	2110	412	343	280	165	63	10	1	2	26	148	295	366
	HDD18.3	4743	670	577	538	413	294	189	114	141	236	402	545	624
	CDD10.0	418	0	0	0	2	27	71	153	121	40	4	0	0
	CDD18.3	10	0	0	0	0	0	1	8	2	0	0	0	0
	CDH23.3	60	0	0	0	0	2	3	42	13	0	0	0	0
	CDH26.7	9	0	0	0	0	0	0	8	1	0	0	0	0
(14) Wind	WSAvg	3.6	4.0	3.7	3.8	3.5	3.5	3.5	3.4	3.2	3.3	3.4	3.7	4.1
(15) Precipitation	PrecAvg	1122	111	103	96	59	57	74	78	91	113	107	106	127
	PrecMax	1427	306	234	268	170	111	138	132	161	280	220	270	278
	PrecMin	746	5	3	26	9	26	7	18	30	34	14	3	27
	PrecStd	156	69	54	63	38	17	28	29	35	63	41	74	59
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.6	8.5	10.9	17.9	22.4	22.9	27.9	25.1	19.8	16.4	11.7	9.4
		MCWB	5.3	5.3	6.3	10.6	14.7	14.5	18.7	16.7	13.3	11.0	8.9	6.0
	2%	DB	6.4	7.1	8.7	12.9	18.7	20.5	23.9	21.6	17.7	14.2	9.0	7.9
		MCWB	4.5	4.5	5.7	8.3	13.3	14.0	17.1	15.4	12.8	10.0	6.8	5.4
	5%	DB	4.8	5.8	7.4	10.7	16.3	18.6	21.5	19.5	16.0	11.7	7.4	6.6
		MCWB	3.2	3.7	4.7	7.0	11.4	13.5	16.6	15.3	12.1	8.6	5.5	4.4
	10%	DB	3.2	4.3	6.1	9.2	14.2	16.7	19.6	17.9	14.7	10.0	5.6	5.0
		MCWB	1.6	2.6	3.8	6.4	10.5	13.1	15.7	14.7	11.9	7.6	4.2	3.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.9	5.7	7.5	10.7	16.1	16.4	19.7	17.7	14.8	12.3	9.5	6.9
		MCDB	7.0	7.6	9.6	16.1	20.3	20.7	24.3	21.7	17.8	14.5	11.3	8.2
	2%	WB	4.7	4.7	6.3	8.8	13.6	15.2	18.3	16.6	13.7	10.5	7.3	5.7
		MCDB	6.2	6.6	8.3	12.0	17.7	18.7	22.6	19.8	15.8	13.6	8.6	7.5
	5%	WB	3.4	3.8	4.9	7.6	12.2	14.3	17.1	15.7	13.0	9.1	5.7	4.5
		MCDB	4.7	5.5	7.1	9.9	15.8	17.3	20.5	18.6	14.9	11.4	7.1	6.3
	10%	WB	1.9	2.7	3.9	6.5	10.8	13.3	16.0	14.9	12.2	7.9	4.3	3.2
		MCDB	3.0	4.2	5.8	8.9	13.8	16.3	19.0	17.6	14.3	9.6	5.6	4.8
(35) Mean Daily Temperature Range	MDBR		6.2	6.7	7.2	7.6	7.5	6.6	6.7	6.9	6.8	6.4	5.3	5.9
	5% DB	MCDBR	6.1	7.1	7.7	10.2	11.1	10.3	10.5	10.1	8.6	8.3	7.3	7.1
		MCWBR	4.4	5.1	5.4	6.3	5.9	5.4	5.1	5.4	4.9	5.4	5.4	5.2
	5% WB	MCDBR	5.4	6.5	7.2	9.0	10.6	8.9	9.6	8.0	7.5	7.2	6.7	6.6
		MCWBR	4.1	4.9	5.3	5.9	6.0	5.2	5.1	4.7	4.7	5.1	5.2	5.1
(40) Clear-Sky Solar Irradiance	taub		0.227	0.288	0.318	0.344	0.347	0.350	0.359	0.347	0.326	0.303	0.368	0.355
	taud		2.051	2.237	2.290	2.359	2.430	2.470	2.469	2.500	2.542	2.455	2.729	2.638
	Ebn at Noon		392	646	777	838	868	872	851	831	780	642	375	218
	Edh at Noon		30	59	83	97	98	97	95	84	66	48	27	18
(44) All-Sky Solar Radiation	RadAvg		0.16	0.68	1.96	3.68	4.68	4.92	4.63	3.56	2.14	0.94	0.24	0.08
	RadStd		0.03	0.11	0.26	0.28	0.27	0.45	0.43	0.37	0.24	0.13	0.04	0.01

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-143	-189	+67	N/A

Nomenclature: See separate page